SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Poster Presentation

Course Code:	**DSA0404**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 – Viva voice Questions**	**Assessment:**	**Assessment Tool 3– Viva voice Questions**
Weightage:	**20% of CIA**	**Total Marks:**	**100**
CO4 Statement:	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Code of Conduct

I Deva Dharshini(192472350) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Deva Dharshini

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

1. What is statistical inference?
Statistical inference is the process of using sample data to draw conclusions about a population. It helps us estimate unknown population values and make decisions based on available data. It mainly involves estimation and hypothesis testing to reach reliable conclusions.

2. Why is statistical inference important in data science?
Statistical inference helps data scientists make decisions using sample data. It is used to estimate population characteristics and test different assumptions. It also helps measure uncertainty and support reliable data-driven conclusions.

3. What is the difference between descriptive statistics and inferential statistics?
Descriptive statistics summarizes and presents the data that has been collected. Inferential statistics uses sample data to make conclusions about a larger population. Descriptive statistics describes existing data, while inferential statistics helps make predictions or decisions about a population.

4. What is a population in statistics?
A population is the complete set of individuals, objects, or observations that are being studied. It can include people, products, customers, machines, or measurements. For example, all customers of a company can be considered a population.

5. What is a sample?
A sample is a smaller subset selected from a population for analysis. It is used to obtain information about the population without studying every individual. For example, 100 products selected from a factory's production can form a sample.

6. Why do data scientists usually work with samples instead of complete populations?
Data scientists usually work with samples because studying an entire population can be expensive, time-consuming, or impossible. A properly selected sample can provide useful information about the population. This makes statistical analysis more practical and efficient.

7. What is a population parameter?
A population parameter is a numerical value that describes a characteristic of the entire population. Examples include the population mean μ, population variance σ^2, and population proportion p. These values are usually unknown and are estimated using sample data.

8. What is a sample statistic?
A sample statistic is a numerical value calculated from the observations in a sample. Examples include the sample mean $\bar{x}$, sample variance s^2, and sample proportion $\hat{p}$. These statistics are commonly used to estimate population parameters.

9. Give a real-world example where statistical inference is used in industry.
A manufacturing company may inspect a sample of products to estimate the defect rate of all products produced. The results from the sample can be used to determine the overall quality of production. This helps the company make quality-control decisions.

10. What are the two major activities involved in statistical inference?
The two major activities involved in statistical inference are estimation and hypothesis testing. Estimation is used to estimate unknown population parameters using sample data. Hypothesis testing is used to determine whether there is sufficient evidence to support or reject a claim.

11. What is the frequentist approach to statistical inference?
The frequentist approach considers population parameters as fixed but unknown values. The sample data is considered random because different samples can produce different results. Statistical conclusions are made using repeated sampling and probability.

12. What is the basic idea behind frequentist statistics?
The basic idea behind frequentist statistics is to use sample data to make conclusions about fixed population parameters. It focuses on the long-run behavior of statistical methods over repeated samples. Confidence intervals and hypothesis tests are commonly used.

13. How does the frequentist approach differ from the Bayesian approach?
In the frequentist approach, the population parameter is considered fixed but unknown. In the Bayesian approach, the parameter is treated as uncertain and represented using a probability distribution. Bayesian methods also combine prior information with observed data.

14. In the frequentist approach, what is considered fixed: the parameter or the data?
In the frequentist approach, the population parameter is considered fixed but unknown. The sample data is considered random because it depends on the sample selected from the population. Therefore, the randomness comes from the sampling process.

15. What does long-run frequency mean in frequentist statistics?
Long-run frequency refers to how often an event occurs when an experiment is repeated many times under the same conditions. For example, a probability of 0.5 means the event is expected to occur about 50% of the time in the long run. It is an important concept in frequentist statistics.

16. Why is random sampling important in the frequentist approach?
Random sampling helps ensure that the sample represents the population fairly. It reduces selection bias and provides a basis for applying probability to the sampling process. This helps make statistical conclusions more reliable.

17. Can a population parameter be considered a random variable in the frequentist approach? Why?
No, a population parameter is not considered a random variable in the frequentist approach. It is treated as a fixed but unknown value. The randomness comes from the sample data and the process used to select the sample.

18. Give an example of a frequentist problem in a data science application.
A data scientist may test whether a new recommendation algorithm increases customer engagement. A sample of users can be analyzed and compared with the existing system. Hypothesis testing can then determine whether the observed improvement is statistically significant.

19. What is a point estimate?
A point estimate is a single numerical value used to estimate an unknown population parameter. For example, the sample mean can be used as a point estimate of the population mean. It provides one best estimate based on the available sample data.

20. What is an estimator?
An estimator is a formula or statistical rule used to estimate an unknown population parameter. It is calculated using information obtained from a sample. For example, the sample mean is an estimator of the population mean.

21. What is the difference between an estimator and an estimate?
An estimator is the formula or method used to estimate a population parameter. An estimate is the actual numerical value obtained after applying that estimator to a particular sample. Therefore, the estimator is the method and the estimate is the resulting value.

22. What is the point estimator for the population mean?
The sample mean $\bar{x}$ is the point estimator for the population mean μ. It is calculated by adding all the sample observations and dividing by the sample size. The formula is .

23. How do you estimate a population proportion from a sample?
A population proportion is estimated using the sample proportion $\hat{p}$. It is calculated by dividing the number of successes by the total number of observations in the sample. The formula is .

24. What is the point estimate of population variance?
The sample variance s^2 is commonly used as the point estimate of population variance σ^2. It measures how much the observations vary around the sample mean. The formula is .

25. Why is the sample mean commonly used to estimate the population mean?
The sample mean is commonly used because it is an unbiased estimator of the population mean. It is also consistent, meaning it becomes closer to the true population mean as the sample size increases. Therefore, it provides a reliable estimate.

26. What properties should a good estimator have?
A good estimator should ideally be unbiased, consistent, and efficient. It should provide values close to the true population parameter and should not vary unnecessarily between samples. These properties make the estimator reliable and useful.

27. What is an unbiased estimator?
An unbiased estimator is an estimator whose expected value is equal to the true population parameter. It does not systematically overestimate or underestimate the parameter. Mathematically, it is represented as .

28. What is meant by the bias of an estimator?
Bias is the difference between the expected value of an estimator and the true population parameter. It shows whether an estimator tends to systematically overestimate or underestimate the parameter. The formula is .

29. What is the difference between bias and variance of an estimator?
Bias measures how far an estimator is systematically from the true population parameter. Variance measures how much the estimator changes from one sample to another. A good estimator generally aims to have low bias and low variance.

30. Why can two different samples produce different point estimates for the same population parameter?
Different random samples contain different observations from the same population. Therefore, their sample statistics can have different values. This natural difference between estimates is known as sampling variability.

31. Why does a point estimate have uncertainty?
A point estimate is calculated from a sample rather than the complete population. Different samples can produce different estimates of the same population parameter. Therefore, the estimate has uncertainty due to sampling variability.

32. What is the sampling distribution of an estimator?
The sampling distribution is the probability distribution of an estimator obtained from repeated samples of the same size. It shows how the estimator varies from one sample to another. It is useful for measuring the uncertainty of an estimate.

33. What is the standard error?
Standard error measures the variability of a sample statistic across repeated samples. It indicates how precisely a statistic estimates the corresponding population parameter. For a sample mean, the standard error is .

34. How is standard error different from standard deviation?
Standard deviation measures the variability of individual observations within a dataset. Standard error measures the variability of a statistic across repeated samples. Therefore, standard deviation describes data spread, while standard error describes estimation uncertainty.

35. How does increasing the sample size affect the standard error of the sample mean?
Increasing the sample size decreases the standard error of the sample mean. This is because standard error is inversely related to the square root of the sample size. Therefore, larger samples generally produce more precise estimates.

36. Why does variability in an estimate decrease when the sample size increases?
A larger sample provides more information about the population and reduces the effect of random sampling differences. As the sample size increases, estimates from different samples become more similar. Therefore, the variability of the estimate decreases.

37. What is the relationship between standard error and confidence intervals?
Standard error measures the uncertainty around a point estimate and affects the margin of error. A larger standard error produces a wider confidence interval, while a smaller standard error produces a narrower interval. The confidence interval is generally calculated as estimate ± critical value × SE.

38. If two samples have different sample means, does it necessarily mean that one sample is incorrect? Explain.
No, different sample means do not necessarily mean that one sample is incorrect. Different random samples naturally contain different observations from the population. Therefore, different sample means are expected because of sampling variability.

39. What is a confidence interval?
A confidence interval is a range of values calculated from sample data to estimate an unknown population parameter. It provides a point estimate along with a measure of uncertainty. For example, an interval of 45 to 55 can be used to estimate a population mean.

40. Why is a confidence interval more informative than a point estimate?
A point estimate provides only one numerical value for an unknown parameter. A confidence interval provides a range of reasonable values along with information about uncertainty. Therefore, it gives a better understanding of the precision of the estimate.

41. What does a 95% confidence level mean in the frequentist approach?
A 95% confidence level means that if we repeatedly take samples and construct intervals using the same method, approximately 95% of those intervals will contain the true population parameter. It describes the long-run reliability of the confidence interval method.

42. What is the difference between a confidence level and a confidence interval?
The confidence level represents the reliability of the interval-producing method, such as 95% or 99%. The confidence interval is the actual range calculated from a particular sample. For example, 95% is the confidence level and (45, 55) is the confidence interval.

43. What factors determine the width of a confidence interval?
The width of a confidence interval mainly depends on sample size, data variability, confidence level, and the critical value. Higher variability and higher confidence levels generally make the interval wider. Larger sample sizes generally make the interval narrower.

44. How does increasing the sample size affect the width of a confidence interval?
Increasing the sample size generally makes the confidence interval narrower. This happens because the standard error decreases as the sample size increases. A narrower interval indicates that the population parameter has been estimated more precisely.

45. How does increasing the confidence level from 95% to 99% affect the confidence interval?
Increasing the confidence level from 95% to 99% makes the confidence interval wider. A higher confidence level requires a larger critical value and therefore a larger margin of error. Thus, higher confidence results in a wider interval.

46. What is the difference between a z-confidence interval and a t-confidence interval?
A z-confidence interval uses the standard normal distribution, while a t-confidence interval uses the t-distribution. The t-distribution is commonly used when the population standard deviation is

unknown, especially for small samples. The t-distribution has heavier tails than the normal distribution.

47. When would you use the t-distribution instead of the normal distribution for estimating a population mean?
The t-distribution is used when the population standard deviation is unknown and the sample standard deviation is used instead. It is especially suitable when the sample size is small. For large samples, the t-distribution becomes very close to the normal distribution.

48. What is hypothesis testing?
Hypothesis testing is a statistical method used to evaluate a claim about a population using sample data. It determines whether there is enough evidence to reject the null hypothesis. The decision is usually made using a significance level such as 5%.

49. What are the null hypothesis (H_0) and alternative hypothesis (H_1)?
The null hypothesis H_0 usually states that there is no significant effect, difference, or relationship. The alternative hypothesis H_1 states that a significant effect, difference, or relationship exists. Sample data is used to determine whether H_0 should be rejected.

50. How can a confidence interval be used to make a decision about a hypothesis?
For a two-sided test, the hypothesized parameter value is compared with the confidence interval. If the hypothesized value lies outside the confidence interval, we reject H_0. If it lies inside the confidence interval, we do not reject H_0.